

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO5: Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Analytical Problem-Solving Assessment

Duration: 1hr

1. **A delivery robot operates in an uncertain environment. Apply the SARSA algorithm to update the action-value function and explain how the learned policy adapts to environmental changes.**

Introduction:

A delivery robot operates in an uncertain environment where it has to navigate from a starting location to a destination while avoiding obstacles, pedestrians, and other unexpected changes. Since the robot may not know the exact outcome of every action in advance, reinforcement learning can be used to learn the best sequence of actions through interaction with the environment. SARSA (State-Action-Reward-State-Action) is an on-policy reinforcement learning algorithm that learns the action-value function $Q(s,a)$ and improves the robot's navigation policy through continuous experience.

SARSA Concept:

SARSA stands for State-Action-Reward-State-Action because the algorithm updates the Q-value using the sequence:

$S \rightarrow A \rightarrow R \rightarrow S' \rightarrow A'$

where:

S = Current state of the robot

A = Current action selected by the robot

R = Reward received after performing the action

S' = Next state reached by the robot

A' = Next action selected in the next state

The important feature of SARSA is that it updates the Q-value using the next action that is actually selected by the current policy. Therefore, SARSA is called an on-policy learning algorithm.

SARSA Update Equation:

$$Q(S,A) \leftarrow Q(S,A) + \alpha [R + \gamma Q(S',A') - Q(S,A)]$$

where:

α = Learning rate, which determines how quickly new information is learned.

γ = Discount factor, which determines the importance of future rewards.

R = Immediate reward received after performing the action.

$Q(S',A')$ = Estimated value of the next state-action pair.

$Q(S,A)$ = Current estimated value of the selected action.

Working of SARSA in a Delivery Robot:

1. Initialization:
The robot initializes the Q-table with zero or small values for all possible state-action pairs.
2. Observe Current State:
The robot observes its current position and environment. The state may include its location, destination, nearby obstacles, battery level, and other environmental information.
3. Select an Action:
The robot selects an action using an ϵ -greedy policy. With probability ϵ , it explores a new action, while with probability $(1-\epsilon)$, it selects the action having the highest Q-value.
4. Perform the Action:
The robot performs the selected action, such as moving forward, backward, left, or right.
5. Receive Reward:
The environment provides a reward based on the result of the action. For example:
 - Reaching the destination $\rightarrow$ large positive reward.
 - Moving closer to the destination $\rightarrow$ positive reward.
 - Taking a longer route $\rightarrow$ small negative reward.
 - Hitting an obstacle $\rightarrow$ large negative reward.
 - Moving into a dangerous area $\rightarrow$ negative reward.
6. Observe Next State:
After performing the action, the robot observes the new state S' .
7. Select Next Action:
The robot selects the next action A' using the same policy.
8. Update Q-Value:
The Q-value is updated using the SARSA equation:
$$Q(S,A) \leftarrow Q(S,A) + \alpha [R + \gamma Q(S',A') - Q(S,A)]$$
9. Repeat:
The above process continues for many episodes until the robot learns an effective navigation policy.

Example:

Consider a delivery robot that has to travel from point A to point B in a warehouse. Initially, the robot does not know which path is safe. It tries different actions and receives rewards from the environment.

If the robot moves toward the destination without hitting an obstacle, it receives a positive reward, increasing the corresponding Q-value. If it hits an obstacle, it receives a negative reward, reducing the Q-value of that action. After repeated trials, the robot learns to select actions that provide higher long-term rewards.

Adaptation to Environmental Changes:

One of the major advantages of SARSA is its ability to adapt to changes in the environment. Suppose an obstacle that was previously absent is suddenly placed on the robot's usual path. When the robot attempts to follow the old route, it receives a negative reward. The corresponding Q-values are updated, and the robot gradually learns to select an alternative route.

Similarly, if a particular route becomes faster or safer, the robot receives higher rewards for using that route, causing its Q-values to increase. Therefore, the learned policy changes continuously based on new experiences.

Advantages of SARSA for Delivery Robots:

- It learns directly from interaction with the environment.
- It considers the actual action selected by the current policy.
- It can handle uncertain and dynamic environments.
- It balances exploration and exploitation using the ϵ -greedy strategy.
- It can learn safer routes because risky actions receive negative rewards.
- It can continuously update its policy when environmental conditions change.

Conclusion:

SARSA is an effective on-policy reinforcement learning algorithm for delivery robot navigation in uncertain and dynamic environments. By repeatedly observing states, selecting actions, receiving rewards, and updating Q-values, the robot gradually learns an optimal navigation policy. Since SARSA learns from the actions actually taken by the robot, it can adapt its behavior when obstacles, traffic conditions, or other environmental factors change. Hence, SARSA enables the delivery robot to achieve safer, more efficient, and adaptive navigation.

Code of Conduct:

I KUSHITHA.G (192425329) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively